# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-POLYFIT-EN · v2.1 · 2026-11-01 |
| Package | star-polyfit |
| Version | 0.1.0 |
| Capability | star.numerics.polyfit |
| Maturity | RELEASED |
| Python | ⩾3.9 |
| Source commit | 94fcf7c |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_polyfit |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_polyfit-0.1.0-py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_polyfit-0.1.0-py3-none-any.whl | 5398a439c1cc6327ed4eb79847e8cf6f49a9734c1bb9f8eb0d10908b1aebbdf9 |
| star_polyfit.py | 06546fdd6df439b32b38da7c47bacd00216407af317f25f0d0e1784605f0f566 |
| star_polyfit-0.1.0-source.zip | 938f1921fc3be5b85b1cb72ba3cb654e5c1d0f6024a078260a9302fa549f1445 |

# 5. What it does, and its limits

Least-squares polynomial of degree 0 to 10, weighted or not: the normal equations are formed and solved in exact rational arithmetic and each coefficient is rounded once, so conditioning costs no digit; exact evaluation and exact residual sum of squares; NIST Wampler1 reproduced exactly; every coefficient of 161 problems equal to SymPy's exact solution.

Exact coefficients are not a good model: a high-degree fit of noisy data oscillates, and the exact coefficients of a badly conditioned problem change a lot when the data change a little; the float nearest to each coefficient is still only a float, and evaluating the fitted polynomial in ordinary float arithmetic far from the origin can lose every digit (use `polyval`, or centre the data). No standard errors, no covariance matrix, no orthogonal polynomials, no automatic choice of the degree, no robust fitting. Exact arithmetic is slow: about a second for 2000 points at degree 10 on a laptop. "Nearest float" refers to the floats passed in, not to the decimal numbers you may have meant.

- `polyfit(xs, ys, degree, weights=None)` returns the coefficients, lowest power first, of the polynomial that minimises the sum of w (y - p(x))². `polyval(coefficients, x)` evaluates a polynomial; `residual_sum(xs, ys, coefficients, weights=None)` returns the weighted sum of squared residuals.
- The normal equations are formed and solved in exact rational arithmetic on the floats given, and each result is rounded once: the conditioning of the problem costs no digit. `polyval` and `residual_sum` are exact sums rounded once as well.
- Measured on 161 problems (120 well conditioned, 40 badly conditioned, and the NIST dataset Wampler1): every coefficient equals the exact rational solution of SymPy rounded to a float and the QR solution of mpmath at 120 digits. NumPy agrees within 1e-9 on the well-conditioned problems; on the badly conditioned ones (x in [1000, 1001], x = 1e6 + k) its coefficients share no digit with the exact ones (the numbers are in the evidence file).

- Every function returns floats or raises `ValueError`: a degree that is not an integer from 0 to 10, data that are not lists or tuples of finite real numbers within 1e100 (booleans refused), different lengths, negative weights, fewer distinct x with a positive weight than degree + 1, a result too large for a float.

## 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

## 7. Citation and bill of materials

```
@software{cavazzini_star_polyfit_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-polyfit}: Least-squares polynomial, correctly rounded},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_polyfit/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

## 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

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